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Technical University of Applied Sciences Würzburg-Schweinfurt
Faculty of Computer Science and Business Information Systems

Master's Thesis

Explainable Causer Attribution for Quality Incidents at BMW Group: Knowledge Graph Link Prediction with Evidence-Based Explanations for Automotive Production

Submitted to the Technical University of Applied Sciences Würzburg-Schweinfurt, Faculty of Computer Science and Business Information Systems, in partial fulfilment of the requirements for the degree of Master of Science (M.Sc.) in Artificial Intelligence

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Abstract

When a defect is detected in automotive production, quality engineers must identify its causer – a supplier or an internal unit – to route the case correctly, for example into the 8D problem-solving process. At BMW Group, this causer attribution today relies on exact-match lookup across defect type, defect place, defect location and material number against historical repair records. This process fails whenever no identical historical case exists, ignores near-miss evidence, and is slow to execute manually, leading to wrong bookings, misdirected 8D processes and supplier disputes. This thesis reformulates causer attribution as an explainable link prediction task on a knowledge graph: given a new incident node, the goal is to predict the missing edge to its causer. A knowledge graph is constructed from BMW's quality notification, inspection and rework data (Qcube/SAP), connecting incidents, reworks and inspections through shared defect, material and cost-centre nodes. Three families of attribution models are compared on this graph: a rule-based baseline that mirrors the current manual process, knowledge graph embeddings (TransE, ComplEx), and a relational graph neural network (R-GCN) that aggregates each incident's neighbourhood for prediction. Every prediction is accompanied by path-based evidence from historical cases, and a locally hosted large language model narrates this evidence in natural language without making predictions itself, keeping the explanation audit-ready for quality engineers. TODO: state the key quantitative result (e.g. Hits@k, MRR or coverage gain over the rule-based baseline) once available. The findings indicate [TODO] and demonstrate [TODO – significance for quality management practice at BMW and, more broadly, for explainable link prediction on event-centric industrial knowledge graphs].

Zusammenfassung

Wird in der Fahrzeugproduktion ein Fehler festgestellt, müssen Qualitätsingenieure den Verursacher – einen Lieferanten oder eine interne Einheit – identifizieren, um den Fall korrekt weiterzuleiten, etwa in den 8D-Problemlösungsprozess. Bei der BMW Group basiert diese Verursacherzuordnung heute auf einem exakten Abgleich von Fehlerart, Fehlerort, Fehlerstelle und Materialnummer mit historischen Reparaturdatensätzen. Dieser Prozess versagt, wenn kein identischer historischer Fall existiert, berücksichtigt keine Fast-Treffer als Hinweis und ist in der manuellen Durchführung langsam – mit der Folge von Fehlbuchungen, falsch adressierten 8D-Prozessen und Lieferantenstreitigkeiten. Diese Arbeit formuliert die Verursacherzuordnung als erklärbare Link-Prediction-Aufgabe auf einem Knowledge Graph: Für einen neuen Vorfallsknoten soll die fehlende Kante zu seinem Verursacher vorhergesagt werden. Ein Knowledge Graph wird aus den Qualitätsmeldungs-, Prüf- und Nacharbeitsdaten (Qcube/SAP) der BMW Group aufgebaut und verbindet Vorfälle, Nacharbeiten und Prüfungen über gemeinsame Fehler-, Material- und Kostenstellenknoten. Drei Modellfamilien zur Verursacherzuordnung werden auf diesem Graphen verglichen: eine regelbasierte Baseline, die den aktuellen manuellen Prozess abbildet, Knowledge-Graph-Embeddings (TransE, ComplEx) sowie ein relationales Graph Neural Network (R-GCN), das die Nachbarschaft jedes Vorfalls für die Vorhersage aggregiert. Jede Vorhersage wird durch pfadbasierte Evidenz aus historischen Fällen begründet; ein lokal betriebenes Sprachmodell verbalisiert diese Evidenz in natürlicher Sprache, ohne selbst Vorhersagen zu treffen, sodass die Erklärung für Qualitätsingenieure prüfbar bleibt. TODO: zentrales quantitatives Ergebnis (z. B. Hits@k, MRR oder Abdeckungsgewinn gegenüber der regelbasierten Baseline) ergänzen, sobald verfügbar.

Acknowledgements

TODO

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1. Introduction

1.1. Motivation

1.1.1. Industry Motivation: BMW Quality Management

In modern automotive manufacturing, maintaining product quality requires the rapid identification and resolution of defects throughout the production process. When a quality incident is detected, one of the first and most critical challenges faced by quality engineers is determining the responsible causer of the defect: specifically, whether the issue originates from a supplier-provided component or from an internal process within production, assembly, logistics, or engineering. This distinction directly impacts the speed and effectiveness of corrective actions. The current BMW quality process follows this principle by forwarding supplier-related incidents to the quality management trend desk, where a quality notification is created in the SAP system and a corresponding supplier complaint and 8D problem-solving process is initiated; internal issues follow an alternative investigation and resolution process.

Despite extensive quality management procedures and the availability of large amounts of historical quality data, accurately identifying the true causer of a defect remains a challenging task. Quality associates often rely on historical incidents, expert knowledge, rework records, inspection information, supplier complaints, and previous 8D reports to support their decisions. However, these data sources are distributed across different systems and are rarely connected in a manner that supports efficient cross-referencing. In practice, such approaches perform adequately when an identical incident has occurred previously, but become significantly less effective when new or slightly different defect scenarios arise. As a result, valuable information contained in related incidents, rework histories, inspections, and historical supplier interactions often remains unused.

The consequences of incorrect causal attribution can be substantial. If a defect is incorrectly classified as supplier-related, an unnecessary supplier complaint and lengthy clarification cycle may be initiated. Conversely, if a genuine supplier issue is not identified at an early stage, corrective actions may be delayed while the underlying defects continue to affect production. Furthermore, the formal confirmation of supplier respon-

sibility often depends on the outcome of subsequent root-cause investigations, which may require considerable time before a definitive acknowledgement becomes available. During this period, the production organization must continue operating under uncertainty, increasing the risk of repeated defects, additional rework activities, and inefficient resource utilization. Reliable causer identification during the earlier stages of defect analysis therefore represents a significant opportunity for improving operational efficiency and reducing problem resolution lead times.

At the same time, the organization possesses an enormous amount of accumulated quality knowledge: historical quality incidents, rework records, inspection records, supplier complaints, root causes, corrective actions, and 8D reports collectively contain valuable information that can support future decisions. However, this knowledge is often fragmented across multiple enterprise systems and remains difficult to reuse systematically. Consequently, quality engineers repeatedly perform similar investigations even though evidence from comparable historical cases may already exist. The challenge is therefore no longer the availability of data, but rather the ability to transform historical enterprise knowledge into actionable and trustworthy decision support.

To address this challenge, this thesis investigates the development of a system capable of identifying likely defect causers based on evidence derived from historical quality knowledge. The proposed solution integrates quality incidents, rework activities, inspection information, supplier-related reports, material data, and historical resolutions into a unified knowledge graph. By utilizing this interconnected knowledge base, the system aims to support quality engineers in determining whether a defect is likely supplier-related or internally caused, identifying recurring patterns, discovering lessons learned from previous incidents, and providing transparent recommendations supported by historical evidence. The objective is not to replace human expertise, but to augment it through explainable and evidence-driven artificial intelligence.

1.1.2. Scientific Motivation

While the industry motivation of this work originates from automotive quality management, the underlying research problem extends far beyond the BMW domain. Similar challenges occur in many complex environments, including aircraft maintenance, semiconductor manufacturing, healthcare incident analysis, industrial fault diagnosis, and large-scale asset management. In all of these settings, organizations possess vast amounts of historical knowledge distributed across heterogeneous data systems, and their decision-makers must continuously evaluate previously unseen situations under uncertainty.

Traditional approaches for supporting such decisions generally fall into three categories. Rule-based systems provide transparent decisions by relying on predefined expert knowl-

edge and deterministic rules; however, their effectiveness is limited by their inability to generalize beyond explicitly defined scenarios, and when no matching rule exists, the system is unable to provide a meaningful recommendation. Machine learning approaches improve predictive performance by learning patterns directly from data, but they often operate as black boxes, making it difficult for domain experts to understand why a specific recommendation was produced; this lack of transparency limits trust and adoption in industrial decision-making environments. Knowledge graph approaches provide a structured representation of enterprise knowledge and support transparent retrieval of related information through explicit relationships between entities; however, most existing applications focus primarily on deterministic retrieval rather than on reasoning over multiple heterogeneous sources of evidence.

This thesis argues that none of these approaches alone is sufficient for supporting complex industrial decision-making tasks. Instead, effective decision support requires a combination of structured enterprise knowledge, evidence retrieval, evidence evaluation, reasoning, and explainability. Consequently, this research proposes an evidence-centric reasoning framework for industrial knowledge graphs. Rather than producing direct predictions from isolated data points, the framework retrieves relevant historical evidence from multiple sources, evaluates the strength and consistency of this evidence, resolves potential conflicts between evidence channels, and generates recommendations through large language model reasoning. The resulting recommendation is accompanied by a traceable explanation that explicitly identifies the evidence supporting the conclusion.

The core research challenge addressed by this thesis is therefore not merely the prediction of a defect's causer. Instead, it investigates a broader question:

> **How can a system support trustworthy decision-making by retrieving, aggregating, evaluating, and reasoning about heterogeneous evidence contained within industrial knowledge graphs?**

To answer this question, the proposed framework integrates symbolic models, graph retrieval, multi-source evidence aggregation and ranking, conflict resolution mechanisms, and large-language-model-based explanation generation. Through this combination, the system seeks to overcome the limitations of traditional rule-based systems, black-box machine learning models, and purely deterministic knowledge-graph retrieval.

The scientific contribution of this work therefore extends beyond the automotive domain. While BMW quality management provides the validation environment, the proposed methodology represents a general framework for explainable decision support on industrial knowledge graphs. The central contribution lies in demonstrating how heterogeneous enterprise knowledge graphs can be transformed into transparent, evidence-based recommendations that remain understandable, auditable, and trustworthy for human

decision-makers.

1.2. Problem Statement and Research Questions

The preceding motivation established why reliable causer attribution matters, both for BMW's quality management process and, more generally, for industrial decision-making under uncertainty. Building on this motivation, this section formalizes the resulting problem and derives the research questions that guide the remainder of this thesis.

Despite the large amount of knowledge generated by quality incidents, rework activities, inspections, supplier complaints, and problem-solving reports, identifying the true causer of a newly detected defect remains a largely manual task. Before corrective actions can be initiated, quality engineers must determine whether a defect originates from a supplier, a production process, an assembly step, or a logistics activity. Current investigation practice relies on expert judgment and exact-match retrieval of previous incidents: effective when an identical case has already been encountered, but prone to failure for incomplete or previously unseen defect scenarios. As a result, relevant evidence scattered across multiple enterprise systems remains underutilized, leading to delayed decisions, repeated investigations, inconsistent causer attribution, and inefficient problem-solving.

Existing technical approaches only partially close this gap. Rule-based systems are transparent but cannot generalize beyond predefined scenarios; machine learning models detect complex patterns but operate as black boxes; and knowledge graphs support structured retrieval but offer limited support for reasoning and decision-making. No existing approach combines historical enterprise knowledge with evidence-based reasoning and explainability into a single decision-support system.

This thesis addresses this gap by investigating an evidence-centric framework that integrates heterogeneous industrial knowledge into a knowledge graph and combines evidence retrieval, multi-source evidence aggregation, and explainable-AI techniques to generate transparent, trustworthy recommendations for defect causer identification. This leads to the following three research questions:

RQ1 – Knowledge Retrieval and Evidence Discovery
How can heterogeneous industrial knowledge sources be integrated and retrieved through a knowledge graph to support the investigation of previously unseen quality incidents?

RQ2 – Evidence Aggregation and Reasoning
How can evidence originating from multiple historical sources be aggregated, evaluated,

and reasoned upon to support reliable defect causer identification and decision-making?

RQ3 – Explainable Decision Support
How can generated recommendations be presented through transparent and auditable evidence-based explanations that support trust and acceptance among industrial domain experts?

1.3. Contributions of the Thesis

This thesis makes the following contributions, corresponding to the research questions introduced in Section 1.2.

Knowledge integration and historical evidence retrieval. In response to RQ1, the thesis develops an approach for integrating heterogeneous industrial quality information and retrieving relevant historical evidence for newly recorded quality incidents. The approach connects information from quality incidents, rework activities, inspection results, and problem-solving records, including situations in which no identical historical case exists.

Multi-source evidence reasoning for causer attribution. In response to RQ2, the thesis develops a method for combining and evaluating evidence from different historical sources. The method distinguishes supporting, conflicting, and insufficient evidence and uses these signals to recommend the most likely process-level causer of a quality incident.

Explainable decision support for industrial users. In response to RQ3, the thesis develops an explainable decision-support framework that presents the recommended causer together with the historical evidence supporting it, relevant contradictions, recurring patterns, applicable lessons learned, and the limitations of the recommendation.

The framework is implemented and evaluated using the distinction between supplier-related and internally caused automotive quality incidents. The first contribution addresses how relevant industrial knowledge can be integrated and retrieved; the second addresses how the retrieved evidence can be evaluated; and the third addresses how the resulting recommendation can be explained and presented for human validation. This produces a clear correspondence between the research questions and the contributions of this thesis, summarized in Table 1.1.

Table 1.1.: Correspondence between the research questions (Section 1.2) and the contributions of this thesis.

Research Question	**Thesis Contribution**
RQ1 – Knowledge Retrieval and Evidence Discovery	Unified industrial knowledge representation and historical evidence retrieval
RQ2 – Evidence Aggregation and Reasoning	Multi-source evidence aggregation and conflict handling
RQ3 – Explainable Decision Support	Transparent causer recommendation with supporting and conflicting evidence

1.4. Thesis Outline

The remainder of this thesis is structured as follows. Chapter 2 introduces the background required to follow the rest of the work: the foundations of knowledge graphs and graph-based machine learning, explainable AI, and the BMW quality management and 8D problem-solving process in which this thesis is situated. Chapter 3 reviews existing approaches to causer and root-cause attribution – rule-based systems, black-box machine learning, and knowledge-graph retrieval – and derives the research gap that motivates this thesis. Chapter 4 formalizes causer attribution as an inductive link-prediction problem on an event-centric knowledge graph and describes the underlying data sources and graph schema. Chapter 5 presents the proposed method: the model architectures compared in this work (rule-based baseline, knowledge graph embeddings, and a relational graph neural network) and the evidence retrieval and explanation generation pipeline built on top of them. Chapter 6 details the experimental setup, including the baselines, the inductive train/test protocol with leakage controls, and the evaluation metrics used to assess predictive performance and explanation faithfulness. Chapter 7 reports the quantitative results, a qualitative analysis of the generated explanations, and the limitations of the proposed approach. Chapter 8 concludes the thesis by revisiting the research questions and contributions in light of the results and outlining directions for future work.

2. Fundamentals

2.1. BMW's Quality Process and Causer Attribution

BMW's quality management process encompasses the detection and documentation of product defects, the containment of their effects, the attribution of responsibility, and the initiation of appropriate problem-solving activities. In the context of this thesis, the central process-level decision is whether a detected defect is attributable to a supplier or to an internal unit, such as assembly, logistics, or another production-related function. This early attribution determines the subsequent investigation and complaint route. The remainder of this section describes the systems, data, and roles involved in this process, from initial defect detection (Section 2.1.1) to the supplier complaint and 8D process (Section 2.1.8). Figure 2.1 summarizes this process at a high level before each stage is described in detail.

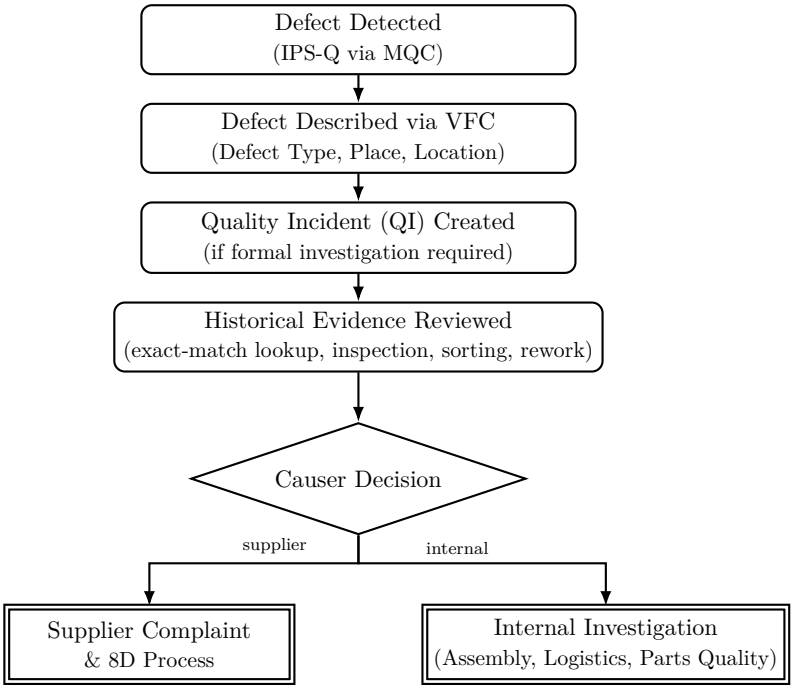


Figure 2.1.: Overview of BMW's current quality-incident and causer-attribution process, synthesized from Sections 2.1.1–2.1.8.

2.1.1. Defect Detection and Recording

The process begins when a defect is detected during production, inspection, or rework. A defect may concern, for example, the appearance, function, condition, installation, or completeness of a component. The **International Production System for Quality (IPS-Q)** provides the production-side context for defects and rework records. Internal material identifies **Modern Quality Capturing (MQC)** as the mobile IPS-Q application used for defect recording and initial causer allocation.

A defect record may contain the affected material, supplier, plant, vehicle, detection time, a textual problem description, and standardized defect characteristics. The IPS-Q-related data fields identified for this process include:

- the material number;
- the supplier number;
- the plant;
- the timestamp;
- the defect description; and
- vehicle identification information.

The quality of the initial textual description is important, because it determines whether comparable historical cases can be found and correctly interpreted. The QMT front-desk process explicitly asks whether the problem description is complete and understandable *before* considering repeat occurrence, causer attribution, rework, potential root causes, and short-term actions. A description such as "door damaged" provides little information for historical comparison, whereas a description such as "damage to the ambient lighting area of the front trim" identifies the affected component and physical position more precisely.

2.1.2. Standardized Fault Coding

To describe defects consistently, BMW uses a standardized fault coding scheme referred to as the **Vehicle Fault Code (VFC; German: *Vereinheitlichte Fehlercodierung*)**. The VFC provides a standardized structure for documenting product-related defects across organizational units, systems, and processes. For this thesis, three VFC elements are particularly relevant: the **defect type**, which describes the observed

form of deviation and effect; the **defect place**, which identifies the affected component or functional area; and the **defect location**, which provides a more precise physical or positional localization. Internal data descriptions contain corresponding fields for defect type, defect place, and defect location, which are used together with material, supplier, plant, and vehicle information to identify defect patterns.

Table 2.1 illustrates these three elements, together with the shared material key, using the door-trim example introduced above.

Table 2.1.: VFC elements and the shared material key, illustrated using the door-trim damage example.

Element	**Description**	**Example Value**
Defect Type	Observed form of deviation and effect	Damage
Defect Place	Affected component or functional area	Door trim, ambient lighting
Defect Location	Precise physical or positional localization	Front right
Material (shared key)	Identifies the affected part	Corresponding door-trim part number

In this thesis, the ordered combination of defect type, defect place, and defect location is referred to as the **defect signature**:

$$D = (T, P, L),$$

where T denotes the defect type, P the defect place, and L the defect location. The defect signature provides a common basis for retrieving related records: the knowledge graph construction described in Chapter 4 likewise treats defect type, defect place, defect location, and material as business keys for linking historical quality incidents and associated rework evidence.

A shared defect signature, however, does not by itself establish an identical causer. The damage in the door-trim example could have been present when the component arrived from the supplier, or it could have occurred during internal transport, assembly handling, or rework. The VFC standard describes what was observed; it does not independently determine how the defect arose or who caused it.

2.1.3. Quality Incident Management

When a defect requires formal investigation, it is documented as a **Quality Incident (QI)**. A QI represents the investigation of a quality problem rather than only an individual defect occurrence. Internal process documentation defines a QI as the formal documentation of a defect detected during production, followed by an investigation to identify the responsible unit. The relevant process documentation associates QI creation, messages, and tasks with QCat and Qcube, systems that also support activities for planning and documenting containment, sorting, rework, and inspection.

The distinction between a **defect occurrence** and a QI is important:

- A *defect occurrence* describes a specific observation on a vehicle or component.
- A *QI* provides the broader investigation context in which one or more observations are examined.

For example, the same door-trim damage may be detected on several vehicles; each vehicle may generate a separate inspection or rework record. These records may be investigated under one QI if they are considered manifestations of the same quality problem. Conversely, two defects with the same VFC description may belong to different QIs if they involve different materials, suppliers, plants, or process conditions. The QI therefore serves as the formal context in which a defect, its possible recurrence, and its potentially responsible units are assessed.

2.1.4. Process-Level Causer Attribution

Process-level causer attribution refers to the identification of the organizational or process-level unit most likely responsible for a quality defect. The initial decision generally distinguishes between a supplier-related issue and an internally caused issue involving assembly, logistics, or another relevant unit. The QMT front-desk process explicitly includes the question of who the causer for the issue is, with assembly, parts quality, and logistics listed as possible areas to be considered.

Causer attribution must be distinguished from **root-cause identification**. Causer attribution determines the organizational or process-level area to which an incident should initially be directed; root-cause identification determines the precise technical or procedural mechanism that produced the defect. For example, assembly may initially be identified as the likely causer before the technical root cause is found to be an unsuitable installation tool or an incorrect handling sequence. Similarly, a supplier may be identified

as the likely responsible party before a manufacturing or packaging deviation is confirmed as the technical root cause.

The QMT front-desk assessment includes more than the defect classification. The documented business questions address:

- whether the problem description is complete;
- whether the issue has occurred previously (repeat occurrence);
- who the likely causer is;
- whether the rework has been fully and correctly recorded;
- which potential root causes and short-term actions are known;
- whether the issue is isolated or part of a series; and
- whether other plants are, or may be, affected.

The result is an early process decision based on the information available at the time. It guides the subsequent investigation but may be revised as additional evidence becomes available.

2.1.5. Historical Lookup and Repeat-Problem Identification

The current investigation process combines structured historical lookup with expert assessment. Internal QMT training material describes a repeat-failure search that queries three different combinations of supplier number, part number, and VFC attributes. An empty result list indicates that no comparable QI was found using the active search configuration; where a QI with the same defect pattern already exists, the new QI can be linked to the original case and marked as a repeat failure.

At a high level, the current process can be summarized as follows:

1. A defect is recorded in IPS-Q through MQC.
2. The defect is described using the VFC and relevant product and process information.
3. A QI is created when formal investigation is required.

4. Historical incidents are searched using configured combinations of supplier, part, and VFC attributes.

5. Related inspection, sorting, and rework information is reviewed.

6. The incident is assigned to a supplier or an internal unit.

7. The responsible expert validates the assignment and initiates the corresponding follow-up process.

This sequence is a synthesis of the documented system landscape and front-desk business questions; the exact execution may vary according to the type and context of the incident. For example, an investigator assessing door-trim damage may retrieve incidents involving the same material and VFC description. If comparable incidents were previously assigned to the supplier, they provide historical support for supplier attribution. Nevertheless, their applicability to the current case must be evaluated using the supplier, plant, detection point, production stage, and other available context.

The current lookup is transparent, because the filter attributes are explicit. Its coverage is nevertheless limited by the defined search combinations and the completeness of the recorded information: a relevant historical case may not be returned if it differs in material, supplier, fault type, or another required attribute. The thesis proposal identifies this exact-match limitation as a central weakness of the existing approach.

2.1.6. Inspection, Sorting, and Rework

Inspection, sorting, and rework are related quality activities, but they serve different purposes and produce different forms of information. **Inspection** evaluates a product's quality characteristics and records the context in which a defect was identified; the inspection information considered in this thesis includes the station, the production stage, the relevant quality check, and the test result. **Sorting** examines a defined population of parts or vehicles, and it distinguishes sorting performed by a supplier, internally by BMW, or through a mixed form involving BMW personnel, a quality service provider, and the supplier. **Rework** documents how a detected defect was corrected; the relevant records may contain the defect type, the repair action (referred to as the rework activity), the affected component, the plant, and the responsible department.

Across the different systems used by QMT, these activities are represented through **Sorting, Rework, and Inspection (SRI) orders**. The documented SRI process includes order creation, instructions, and the recording of inspection results as OK or NOK (not OK).

These activities provide complementary perspectives: inspection records where and under which conditions a defect is detected, sorting records the result of examining a defined population, and rework records how the defect was corrected. These perspectives must not be confused with causation. The place at which a defect is detected is not necessarily the place at which it was created; likewise, the unit that performs the repair is not automatically the causer. For example, door-trim damage detected at an assembly inspection point may already have existed before the component entered assembly; conversely, a conforming component may be damaged during installation and subsequently reworked at another station. Detection, correction, and causation therefore remain separate aspects of the investigation.

2.1.7. Roles in the Quality Process

Several roles contribute to defect detection, investigation, containment, and problem-solving:

Production and Assembly
Identify defects in the manufacturing environment and provide the initial defect and process context. MQC is positioned in the internal ecosystem as the application for recording a defect and performing an initial allocation to a likely causer.

Quality Engineers and QMT Front-Desk Personnel
Examine the problem description, compare the issue with previous cases, assess a likely causer, and determine the appropriate follow-up process. Their documented business questions include repeat occurrence, causer identification, rework completeness, potential root causes, short-term actions, and affected plants.

Internal Units (Assembly, Logistics, Parts Quality, and Related Units)
May receive incidents for further investigation when internal responsibility is suspected. These units appear as potential causer categories in the processing paths documented in the QMT front-desk material.

Suppliers
Participate in the complaint and problem-solving process when supplier responsibility is suspected. A dedicated supplier portal, qWin, supports the exchange and management of supplier-related 8D information, including requested 8D reports. External quality service providers may perform sorting activities, while mixed sorting additionally involves the supplier.

2.1.8. Supplier Complaints and the 8D Process

If a defect is considered supplier-related, the incident can proceed to a supplier complaint process. The documented system landscape connects the quality notification to **8D (eight-disciplines) reporting**.

The 8D method is a structured problem-solving approach used to identify, correct, and prevent recurring product or process problems. It progresses from team formation and problem definition, through containment and root-cause analysis, to permanent corrective actions and the prevention of recurrence.

The 8D method originated in the automotive industry as a standardized framework for supplier and internal problem-solving and is applied within BMW's quality management process whenever a defect is confirmed as supplier-related. Although the disciplines are numbered sequentially, they are not always executed strictly one after another in practice: D3 (interim containment) is typically initiated in parallel with, or even before, the completion of D4 (root-cause analysis), since protecting the customer and the production process cannot wait until the root cause is fully understood.

In practice, each QI is assigned a priority level, and the eight disciplines are governed by corresponding target timelines that differ between high- and low-priority cases.

The cross-functional team formed under D1 typically draws on the roles introduced in Section 2.1.7: the responsible quality engineer coordinates the investigation, while representatives from production, assembly, and, where relevant, the supplier contribute the domain-specific knowledge needed to describe the problem, contain it, and identify its root cause.

The 8D reports generated through this process also complement the historical lookup described in Section 2.1.5: whereas that lookup retrieves comparable past QIs based on supplier, part, and VFC attributes, an attached 8D report reveals not only that a comparable incident existed, but how it was investigated and resolved, and through which discipline-specific findings.

Each discipline is documented with a responsible owner, supporting evidence, and a completion date. This staged documentation means that an 8D report's evidentiary value depends on how far the investigation had progressed at the time it is reviewed: a report that is still open may contain only a preliminary problem description and containment action, whereas a closed report additionally documents a verified root cause and validated corrective actions.

For the purposes of this thesis, this discipline structure matters because an 8D report is not a single, undifferentiated text block: each discipline corresponds to a distinct, dated

piece of evidence, such as a problem description, a containment action, or a verified root cause, that can in principle be linked to the underlying QI as a separate item of evidence rather than treated as one opaque document. Table 2.2 summarizes the eight disciplines reviewed for this thesis.

Table 2.2.: The eight disciplines (8D) of the problem-solving process.

Discipline	**Title**	**Purpose**
D1	Establish the Team	Form a cross-functional team with the product and process knowledge required to investigate and resolve the problem.
D2	Define and Describe the Problem	Describe the problem clearly and, where possible, quantitatively. A description should establish what occurred, where and when it occurred, who or what was affected, and the extent of the problem.
D3	Develop and Implement Interim Containment Actions	Introduce temporary measures to isolate the problem and protect the customer or production process while the root cause is being investigated.
D4	Determine and Verify Root Causes	Identify and verify the causes that explain why the problem occurred. This discipline also considers why the defect was not detected earlier. Methods such as the 5 Whys and cause-and-effect diagrams may support the analysis.
D5	Select and Verify Permanent Corrective Actions	Identify permanent actions that address the verified root cause, and confirm that the selected actions are capable of resolving the problem.
D6	Implement and Validate Corrective Actions	Implement the selected permanent corrective actions and verify their effectiveness using appropriate evidence.
D7	Prevent Recurrence	Modify relevant systems, procedures, practices, or controls to prevent the same or similar problems from recurring.
D8	Recognize the Team and Close the Process	Recognize the team's contribution and formally close out the problem-solving process.

For causer attribution, the 8D process is particularly relevant because its disciplines produce different forms of historical evidence: D2 contains a structured description of the problem, D3 records the interim containment measures, D4 documents the verified root cause, and D5 through D7 describe the selected, implemented, and preventive corrective actions.

Completed 8D reports can therefore provide evidence about how comparable historical incidents were investigated and resolved. However, a historical 8D report should not automatically determine the causer of a new defect; its relevance depends on whether the current and historical cases are comparable in terms of defect, material, supplier, plant, and process context. For example, an earlier 8D report may document that comparable door-trim damage was related to supplier packaging. This provides relevant historical evidence if the current incident involves the same material, supplier, and incoming inspection point; its relevance is weak if the defect was first identified only after an internal assembly activity. Within this thesis, 8D information is consequently treated as an important historical evidence source rather than as an automatic attribution rule.

2.1.9. Summary

This section has described BMW's current quality process and the data it produces. A defect is first recorded in IPS-Q through MQC and described using the VFC, whose defect type, defect place, and defect location form the defect signature introduced in Section 2.1.2; when formal investigation is required, this is escalated to a QI, which may bundle several defect occurrences or, conversely, split identical-looking defects across several QIs depending on material, supplier, plant, and process context (Section 2.1.3). The central decision at this stage is process-level causer attribution, which must be kept distinct from technical root-cause identification and which the QMT front-desk process supports through a fixed set of documented business questions (Section 2.1.4). Today, this decision relies on a structured but exact-match historical lookup across supplier, part, and VFC attributes, which is transparent but fails whenever no identical historical case exists (Section 2.1.5). Inspection, sorting, and rework records provide complementary evidence about where a defect was detected and how it was corrected, but none of these, nor the unit that performs them, can be equated with causation (Section 2.1.6). Several roles, from production and assembly to quality engineers, internal units, and suppliers, contribute evidence at different stages of this process (Section 2.1.7), and supplier-related incidents additionally generate 8D reports, whose eight disciplines yield further, differently reliable forms of historical evidence depending on how far an investigation has progressed (Section 2.1.8).

Across all eight subsections, the same limitation recurs: relevant evidence is distributed across IPS-Q, QCat, Qcube, SRI orders, and 8D reports, and today it is combined only through exact-match retrieval on a small set of attributes, rather than through systematic reasoning over all available historical evidence. This motivates the knowledge-graph-based approach developed in the remainder of this thesis, which integrates these heterogeneous data sources into a single structure (Chapter 4) and supports evidence retrieval, aggregation, and explanation beyond exact-match lookup (Chapter 5).

2.2. Knowledge Graphs and Link Prediction

A knowledge graph represents knowledge as a network of entities and their relationships. Entities correspond to distinguishable real-world objects or abstract concepts and are represented as nodes, while relations are represented as typed edges connecting these nodes. Unlike conventional data representations that treat records independently, a knowledge graph explicitly captures how different objects relate to one another, providing a structured and semantically meaningful representation of interconnected information [1].

Formally, a knowledge graph can be defined as a collection of facts represented as triples

$$(h, r, t),$$

where h denotes the head entity, r the relation, and t the tail entity [1, 2]. The relation specifies the meaning of the connection between the two entities. For example, the following triples illustrate this structure:

$$\text{(Paris, CAPITAL_OF, France)}$$
$$\text{(Component, SUPPLIED_BY, Supplier)}$$
$$\text{(Incident, HAS_DEFECT, Defect)}$$

The first states that Paris is the capital of France; the second, that a component originates from a particular supplier; and the third records the association between a reported incident and its observed defect.

A knowledge graph is formed by combining many such triples, allowing the same entity to participate in multiple relations. A single component, for instance, may simultaneously be connected to a supplier, a production plant, a recorded incident, and an observed defect. The resulting structure is therefore a multi-relational graph, in which different edge types express different semantic relationships [2]. Entities and relations may additionally carry types and properties: an incident entity may be described by attributes such as an identifier, a creation date, and a status, while a defect entity may be described by its type, location, and affected area. These descriptive properties characterize individual entities, whereas relations express how entities are connected; together, they allow heterogeneous information to be represented within a common structure while preserving its original domain-specific meaning [1].

This representation is particularly suitable for domains in which relevant information is distributed across multiple sources. In industrial quality management, for example, analyzing a reported incident may require information about defects, materials, plants, suppliers, inspections, rework, and containment actions. These objects describe different aspects of the same underlying context; representing them as connected entities makes

it possible to retrieve relevant information through their relationships, rather than examining each source independently. The same principle applies more broadly: wherever an organization's knowledge is distributed across heterogeneous records that share common objects, a knowledge graph offers a structured way to reconnect that information for retrieval and reasoning.

2.2.1. Knowledge Graph Completion and Link Prediction

Knowledge graphs are generally incomplete, because not every valid relationship is observed or explicitly recorded. The absence of a link does not necessarily imply that no relationship exists; it may instead indicate that the corresponding fact has not yet been identified, entered, or derived from the available information. Knowledge graph completion addresses this incompleteness by identifying plausible missing facts from the existing graph structure [1].

Link prediction is one of the principal tasks used for this purpose. Given an incomplete triple, link prediction estimates the missing entity or evaluates whether a proposed relationship is plausible [2, 3]. The task may take different forms:

$(h, r, ?)$ the tail entity is unknown,
$(?, r, t)$ the head entity is unknown,
(h, r, t) the plausibility of a complete triple is evaluated.

For example, given the incomplete triple

$$(\text{Incident}, \text{CAUSED_BY}, ?),$$

the link prediction task is to identify the most plausible causer among a set of candidate entities. Candidates are typically ranked according to a scoring function that estimates the plausibility of each possible completion; the highest-ranked candidate is the entity considered most compatible with the learned graph patterns. Link prediction therefore transforms the graph completion problem into a structured prediction task over entities and relations [3, 2].

Knowledge graph embedding methods commonly support this task by mapping entities and relations into a continuous vector space. A scoring function then evaluates how plausible a given combination of head entity, relation, and tail entity is; the representations are learned such that observed triples receive higher plausibility scores than incorrect or unobserved combinations. However, the ability of such a model to handle new entities depends on whether it follows a transductive or an inductive learning setting.

2.2.2. Transductive and Inductive Link Prediction

In a **transductive** setting, the set of entities considered during inference is already known at training time: the model observes a graph containing all relevant entities and learns representations for them, even if some of their relationships are hidden and later used for evaluation. Conventional knowledge graph embedding methods frequently follow this paradigm by assigning a trainable embedding vector to each entity in the training graph [2, 4]. Such models can predict missing links between entities for which embeddings have already been learned; for example, if an incident and all of its possible causers were included in the training graph, the model could estimate which of the missing relations between these known entities is most plausible. This setting, however, assumes that the entities required during inference were already available at training time: an entity-specific embedding cannot be retrieved for an entity that did not occur in the training graph [4, 3].

This limitation is significant in dynamic application environments, where new products, incidents, materials, or other business objects may continuously be added after a model has been trained. Retraining the entire model whenever a new entity appears is generally undesirable, and, more importantly, a model that depends only on memorized entity representations does not demonstrate that it has learned relational patterns transferable to previously unseen cases.

Inductive link prediction addresses these limitations by requiring a model to generalize to entities that were not part of the training graph [4, 5]. During inference, the graph may contain entities absent from training; instead of relying exclusively on a stored embedding for every entity, an inductive method derives information from transferable features, relations, or structural patterns in an entity's graph neighborhood. A neighborhood describes the entities and relations surrounding a target entity; for an incident, this neighborhood may include its associated defect, material, plant, and inspection or rework records. Although the incident itself is new, some of these neighboring entity types and relational structures may resemble patterns observed in historical incidents, and an inductive method can use the structure surrounding the new entity to support its prediction. Subgraph-based inductive approaches formalize this idea by extracting a local subgraph around the candidate entities and learning the relational and structural patterns within such subgraphs, rather than only the identities of individual entities [4]; the learned patterns can then be applied to a new subgraph containing an entity that was absent during training.

The distinction can be summarized as follows: transductive link prediction predicts missing relations among entities that are already known during training, whereas inductive link prediction transfers learned relational patterns to entities that were not present during training. This distinction concerns the availability of *entities*, rather than only the presence or absence of individual edges: hiding a relation for an incident that was

already observed during training remains a transductive task, whereas predicting the same type of relation for an incident that was entirely absent from training constitutes an inductive task.

2.2.3. Relevance to the Proposed Methodology

RQ1 (Section 1.2) asks how heterogeneous industrial knowledge can be retrieved through a knowledge graph to support the investigation of *previously unseen* quality incidents. The concepts introduced above give this question a precise formulation: causer attribution is framed as the prediction of a missing relation between a quality incident and a candidate causer,

$$(q, \text{CAUSED_BY}, c),$$

where q denotes a quality incident and c a candidate causer. Historically resolved incidents provide examples for which this relation is already known; a newly created incident has no known causer and must instead serve as the head entity of an incomplete triple,

$$(q_{\text{new}}, \text{CAUSED_BY}, ?).$$

Because q_{new} did not exist when the historical training graph was built, no entity-specific representation could have been learned for it – this is precisely the inductive setting described in Section 2.2.2. What remains available is q_{new}'s local graph context: its connections to material, plant, inspection, and other quality information. Rather than depending on the identity of the incident, the methodology developed in this thesis compares this relational neighborhood against the structures of previously resolved cases, directly operationalizing RQ1's requirement to retrieve evidence for incidents that have no exact historical match.

This inductive framing is also what connects the proposed methodology to the broader scientific motivation outlined in Section 1.1.2: treating causer attribution as link prediction under an open, continuously growing entity set, rather than a fixed, previously observed one, is itself the methodological contribution that generalizes beyond BMW's quality process to any industrial decision-support setting in which an organization's knowledge graph keeps growing after deployment. Chapter 6 reflects this requirement directly: evaluation holds out incidents whose entities are entirely unseen during training, since a transductive evaluation would not reflect this intended, continuously evolving setting.

2.3. Explainable Artificial Intelligence

TODO

3. Related Work

3.1. Root-Cause and Causer Attribution in Manufacturing Quality

TODO

3.2. Research Gap

TODO

4. Problem Formulation and Data

4.1. Formal Problem Definition

TODO

4.2. Data Sources and Graph Schema

TODO

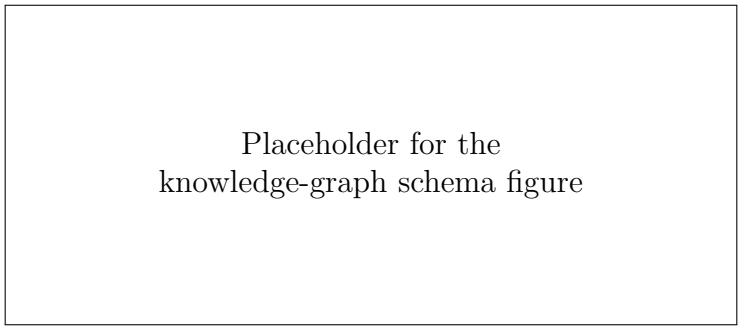


Figure 4.1.: Knowledge-graph schema (placeholder).

As shown in Figure 4.1, TODO.

5. Method

5.1. Model Architecture

TODO

5.2. Explanation Generation

TODO

```
// Retrieve candidate causers for an incident
MATCH (i:Incident {id:
    $incidentId})-[:AFFECTS]->(p:Part)<-[:SUPPLIES]-(s:Supplier)
RETURN s.name AS candidate, count(p) AS evidence
ORDER BY evidence DESC
LIMIT 5
```

Listing 5.1: Example graph query (placeholder).

6. Experimental Setup

6.1. Baselines

TODO

6.2. Evaluation Metrics

TODO

7. Results and Discussion

7.1. Quantitative Results

TODO

Table 7.1.: Comparison with baselines (placeholder values).

Method	Top-1 Acc.	Top-3 Acc.	MRR
Baseline A	–	–	–
Baseline B	–	–	–
Proposed method	–	–	–

7.2. Qualitative Analysis of Explanations

TODO

7.3. Limitations and Threats to Validity

TODO

8. Conclusion and Future Work

8.1. Conclusion

TODO: Answer each research question explicitly.

8.2. Future Work

TODO

A. Additional Tables

Table A.1.: Overview of answer types (placeholder).

Heading 1	**Heading 2**
Title 1	Description 1
Title 2	Description 2

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Bibliography

[1] Shaoxiong Ji et al. “A Survey on Knowledge Graphs: Representation, Acquisition, and Applications”. In: *IEEE Transactions on Neural Networks and Learning Systems* 33.2 (2022), pp. 494–514. DOI: 10.1109/TNNLS.2021.3070843.

[2] Quan Wang et al. “Knowledge Graph Embedding: A Survey of Approaches and Applications”. In: *IEEE Transactions on Knowledge and Data Engineering* 29.12 (2017), pp. 2724–2743. DOI: 10.1109/TKDE.2017.2754499.

[3] Andrea Rossi et al. “Knowledge Graph Embedding for Link Prediction: A Comparative Analysis”. In: *ACM Transactions on Knowledge Discovery from Data* 15.2 (2021), pp. 1–49. DOI: 10.1145/3424672.

[4] Komal K. Teru, Etienne G. Denis, and William L. Hamilton. “Inductive Relation Prediction by Subgraph Reasoning”. In: *Proceedings of the 37th International Conference on Machine Learning*. Vol. 119. Proceedings of Machine Learning Research. 2020, pp. 9448–9457.

[5] Mikhail Galkin, Max Berrendorf, and Charles Tapley Hoyt. “An Open Challenge for Inductive Link Prediction on Knowledge Graphs”. In: *Workshop on Graph Learning Benchmarks, The Web Conference (WWW)*. 2022. arXiv: 2203.01520.

Declaration of Authorship

I hereby declare that I have written the present Master's Thesis independently and have not submitted it elsewhere for examination purposes. All sources and aids used have been indicated, and all direct and indirect quotations have been marked as such.

Riya Biju, on October 23, 2026

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